Robust Epileptic Seizure Detection Based on Biomedical Signals Using an Advanced Multi-View Deep Feature Learning Approach

Ijaz Ahmad, Zhenzhen Liu, Lin Li, Inam Ullah, Sunday Timothy Aboyeji, Xin Wang, Oluwarotimi Williams Samuel, Guanglin Li, Yuan Tao, Yan Chen, and Shixiong Chen

***Abstract*—Epilepsy is a neurological disorder characterized by abnormal neuronal discharges that manifest in life-threatening seizures. These are often monitored via EEG signals, a key aspect of biomedical signal processing (BSP). Accurate epileptic seizure (ES) detection significantly depends on the precise identification of key EEG features, which requires a deep understanding of the data's intrinsic domain. Therefore, this study presents an Advanced Multi-View Deep Feature Learning (AMV-DFL) framework based on machine learning (ML) technology to enhance the detection of relevant EEG signal features for ES. Our method initially applies a fast Fourier transform (FFT) on EEG data for traditional frequency domain feature (TFD-F) extraction and directly incorporates time domain (TD) features from the raw EEG signals, establishing a comprehensive traditional multi-view feature (TMV-F). Deep features are subsequently extracted autonomously from optimal layers of one-dimensional convolutional neural networks (1D CNN), resulting in multi-view deep features (MV-DF) integrating both time and frequency domains. A multi-view forest (MV-F) is an interpretable rule-based advanced ML classifier used to construct a robust, generalized classification. Tree-based SHAP explainable artificial intelligence (T-XAI) is incorporated for interpreting and explaining the underlying rules. Experimental results confirm our method's superiority, surpassing models using TMV-FL and single-view deep features (SV-DF) by 4% and outperforming other state-of-the-art methods by an average of 3% in classification accuracy. The AMV-DFL approach aids clinicians in identifying EEG features indicative of ES, potentially discovering novel biomarkers, and improving diagnostic capabilities in epilepsy management.**

***Index Terms*—Machine learning, biomedical signal processing, EEG, explainable AI, epileptic seizure diagnosis.**

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I. Introduction

EPILEPSY is a non-communicable disorder marked by seizures due to abnormal activity of the brain [1]. These seizures, manifesting as focal or generalized episodes, evoke a range of symptoms, including transient loss of consciousness, muscle contractions, and changes in emotional or behavioral states [2]. As reported by the World Health Organization (WHO), epilepsy affects over 50 million individuals globally, constituting approximately 1% of the world's population [3]. Electroencephalography (EEG) is a non-invasive biomedical signal processing (BSP) tool frequently used for epilepsy monitoring and diagnosis, which involves the attachment of multiple electrodes to the patient's scalp [4], [5]. The recordings yield valuable insights into brain electrical activity, facilitating the detection of indicative seizures [6]. Traditionally, neurologists perform visual EEG inspection for epileptic seizure (ES) detection, an approach that is laborious, time-consuming, costly, and susceptible to human error. Hence, automatic detection systems can substantially enhance the diagnostic process for clinicians. Advancements in machine learning (ML) have facilitated the integration of intelligent algorithms to enhance EEG-based ES detection performance. These algorithms employ efficient ML model techniques such as random forest classifiers (RF) [7], support vector machines (SVM) [8], k-nearest neighbors (KNN) [9], and decision tree-based classifiers [10], using time and frequency domain features [10]. They utilize feature extraction methods like fast Fourier transforms (FFT) and wavelet transforms (WT) [11]. Despite the development of numerous feature extraction and classification approaches, extracting useful features for accurate ES remains a formidable challenge. Deep learning (DL) has emerged as an important ML paradigm in recent years, garnering significant interest in feature learning in EEG ES detection [12].

Furthermore, multi-view learning techniques (MV-LT) have been increasingly applied in the field of ES detection [13], [14], [15]. The technology is categorized into three primary types: mutual training, multi-kernel learning, and dimensional reduction learning. Mutual training or co-training enhances data consistency among views; multiple kernel learning combines different kernels to boost learning; and dimensional reduction learning or subspace learning uses common subspaces to understand multi-view data. Despite their differences, all these approaches fundamentally rely on principles of consensus or complementarity. Additionally, to explain the multi-view learning technology (MV-LT), explainable artificial intelligence (XAI) is employed to elucidate, interpret, and identify the essential features contributing to the output of the models, particularly in ES detection. XAI is gaining prominence in healthcare, including ES detection, as it enables clinicians to comprehend artificial intelligence decision-making processes and identify possible errors or biases in predictions [16]. XAI is used in EEG ES to analyze the features based on the model outcomes and highlight the important features that have the best performance [17].

On the other hand, existing literature predominantly focuses on traditional features derived from EEG biomedical signal data for accurate ES detection. However, the performance of the classifiers heavily depends on the user's selection of the features, necessitating feature extraction methods that are independent of user input or an in-depth understanding of EEG data. Furthermore, deep learning models and their features are challenging to interpret, thereby limiting their application in clinical settings. Traditional methods that incorporate expert decisions and rules into the model may find it difficult to handle large-scale, high-dimensional EEG datasets. To address the informative challenge and construct effective EEG features for ES detection, the proposed study introduces an advanced multi-view deep feature learning (AMV-DFL) approach. The proposed AMV-DFL framework employed in this study can be summarized as follows:

- *Traditional multi-view features* are extracted from the raw EEG data in time domain signals, while fast Fourier transform (FFT) is applied for frequency domain signals in the initial stages of the framework. Subsequently, the extracted features are fed into a one-dimensional convolutional neural network (1D CNN) model. The procedure effectively enables the model to capture and analyze time and frequency features from the initial EEG data.
- *Construction of multi-domain deep feature extraction* to enhance the efficacy of deep features. The 1D CNN constructs multi-domain deep features utilizing the TMV-F data. The resulting multi-domain deep features (MV-DF) exhibit multi-view information regarding both time and frequency domains.
- *Construction and generate rules-based classifiers* Using a deep feature learning approach, a multi-view forest classifier (MV-F) is employed to construct and perform classifications. The method generates rules for non-seizure (NS), transition (TS), and seizure (SZ) classes, upon which the proposed approach performs classification. According to the authors' knowledge, this is the first study, to introduce rules-based classification in EEG epileptic seizure detection. The resulting classifier offers a more generalized solution for EEG signal-based ES detection.
- *Construction of interpretability and explainability* of the rules-based model using tree-based explainable artificial intelligence (T-XAI) in EEG ES detection, encouraging enhanced comprehension and trust in the model decision-making processes. Furthermore, the approach explains the significant deep features contributing to rule generation.

The proposed framework offers several advantages, as it extracts TMV-F from EEG signal data and trains the base classifier (ID CNN) while combining the deep features from the base classifier of different domains, which are then used to construct rules for each class using a multi-view forest classifier. This approach effectively optimizes feature representation for ES detection, resulting in a more robust and versatile solution for the NS, TS, and SZ states of the EEG.

The remaining study is organized as follows: Section II discusses the related work. Section III presents the details of the proposed framework. Section IV describes the experimental results and discussion. Finally, the conclusion and exploration of future work are in Section V.

II. Related Work

Epileptic seizure (ES) detection often relies on EEG signals, which reflect the brain neurons' activities. Converting the EEG signal information into distinct outputs helps classify various epilepsy states. Recently, different epilepsy detection algorithms have been used for classifying epilepsy status using EEG. For instance, multi-view learning techniques (MV-L) were applied for automatic epilepsy classification [21], [22], while both multi-view learning (MV-L) and deep learning (DL) approaches separately have shown good performances, but their integration could further enhance ES detection performance.

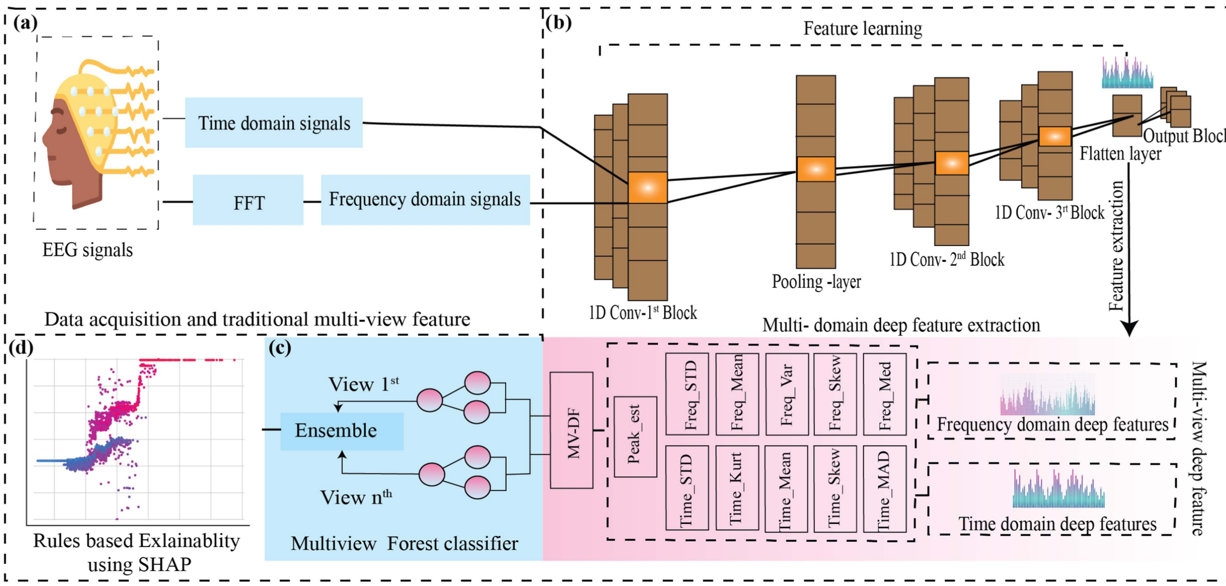

Fig. 1. The advanced multi-view deep feature learning framework for EEG epileptic seizure detection. Besides, the figures in panels (a), (b), (c), and (d) represent the different components of the proposed approach.

A. Epileptic Seizure Detection Using Machine Learning Models

In recent decades, machine learning (ML) has gained much importance in EEG-based ES detection. In [18] implemented four approaches: decision tree (DT), random forest (RF), support vector machine (SVM), and a hybrid of SVM-DT and SVM+RF yielding superior results with 98.10% accuracy. In study of [19] processed the Bonn EEG dataset by normalizing and dividing it into one-second segments before applying a four-level discrete wavelet transform (DWT). The study compared various models, including logistic regression, KNN, SVM, RF, artificial neural network (ANN), and CNN models, while convolutional neural networks (CNN) achieved the highest prediction accuracy. CNNs have been used in different ways for processing EEG signals, including the application of one-dimensional convolution on raw EEG data for ES prediction and encoding EEG signals into pixel colors to create two-dimensional patterns for ES prediction [20].

B. Multi-View Feature Learning (MV-FL)

MV-FL is an advanced machine learning technology where data is represented by multiple distinct feature views. Multi-view learning (MV-L) captures complementary information available in different views of the input data, ultimately improving performance across various applications [21]. Indeed, multi-view learning techniques are applied to ES detection using EEG signals [21], [22]. Approaches like tensor decomposition have been employed on multi-view features to extract new attributes, enhancing the classification performance for ES detection [23]. In the proposed study, a multi-view forest classifier is applied for various object detection tasks, often leading to improved prediction accuracy due to the utilization of diverse feature sets and perspectives on the same data.

III. The Proposed Framework for Epileptic Seizure Detection Using EEG Signals

In this section, we explain an advanced multi-view deep feature learning framework (AMV-DFL), including TMV-F and MV-DF, rule generation, explanation, and interpretation using tree-based explainable artificial intelligence (T-XAI), as shown in Fig. 1.

A. Construction of Multi-View Feature (TM-VF)

1) Time and Frequency Domain Signals: The EEG signals always represent features in the time domain. However, to convert the time-domain signals into the frequency domain, we employ transformation techniques. The fast Fourier transform is commonly used to convert time-domain signals into frequency-domain signals, offering an alternative representation with the EEG's short duration. The method yields highly precise frequency characteristics. The approach not only enhances detection accuracy by leveraging the inherent information in the frequency domain but also aids in better understanding the underlying neurological phenomena associated with epilepsy. The proposed methods extract features from time and frequency from the raw EEG signals, creating comprehensive, multi-view EEG signal data. This enhancement increases the robustness of our analysis. EEG signals are essentially time-dependent sequences that offer each discrete sample a snapshot of energy intensity at a particular time. The proposed study presents the intrinsic signals as time-domain features. Fig. 2(a), (b) illustrates an EEG channel in the time domain (TD) and frequency domain (FD), where the time domain, the x-axis shows the time maximum of 23.7 sec and the y-axis denotes the signal amplitude, while in the frequency domain, the x-axis presents the frequency (Hz) and the vertical axis represents the signal amplitude (μ). Suppose the FFT uses the sequence of N numbers as signal data points

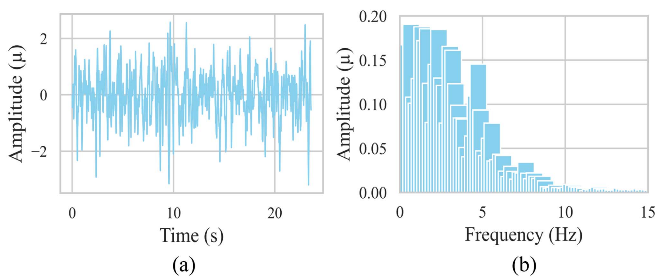

Fig. 2. Comparative analysis of EEG signals in both the time and frequency domains. Sub-figure (a) shows the time domain representation of the EEG signal after filtering, where the x-axis represents time in seconds and the y-axis indicates the signal amplitude μV. The sub-figure (b) provides the frequency domain representation of the same EEG signal after filtering. The x-axis denotes the frequency in Hz, and the y-axis corresponds to the amplitude.

TABLE I
THE PARAMETERS OF THE PROPOSED FEATURE EXTRACTION ARCHITECTURE OF THE BASE MODEL

Layer	Nodes	Kernel Size	Output Shape	Parameters
Conv1D_1	24	5	510 × 24	96
MaxPooling1D	-	2	255 × 24	-
Conv1D_2	1	3	253 × 16	1168
Conv1D_3	8	3	251 × 8	392
Flatten	-	-	2008	-
Dense	3	-	3	40180

of frequency $x_0, x_1, \ldots, x_{(N-1)}$ is calculated as:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-2\pi i \frac{kn}{N}} \tag{1}$$

where $X(k)$ is the k^{th} coefficient in the frequency domain, and $x(n)$ is the n^{th} data point in the time domain sequence. After extracting TMV-F, the time and frequency domain signals are fed into the 1D CNN model to train for deep feature extraction, which is discussed in the next section.

B. Construction of Multi-Domain Deep Feature Extraction

1) Deep Feature Extraction From the Base Model: After performing traditional time and frequency domain feature extraction, a one-dimensional convolutional neural network (1D CNN) as the base model is utilized. The model has shown exceptional performance in extracting efficient spatial features [10]. The base mode (1D CNN) configuration, including the number of parameters, kernel size, and number of layers, has been iteratively optimized to maximize accuracy. The specific parameters are summarized in Table I which are the same for both the time and frequency domain deep feature extraction processes. The base model uses time and frequency domain signals as multi-view and automatically extracts a set of high-level and low-level features known as deep features (DP). The architecture of the base classifier consists of three convolutional layers (CL) with batch normalization and activation function (ReLU), one max-pooling layer (MAX), a fully connected (FC), and a flatten layer. The first block of the convolution network is designed to learn high-level features from the time and frequency domain signals that have kernel sizes of 5 and a filter of 24 with the input shape (512, 1), each producing a feature map that represents the kernel's response to the input data. After the first Block of CL, an MAX layer with a 2 kernel size is employed to reduce computational costs. The last two CL blocks aim to learn the low-level deep features that have 'padding =same' and a kernel size of 3, while the filter is 16 and 8. The rectified linear unit (ReLU) is used as the activation function in the 1D CNN to rescale the negative values of the feature produced by the noise and artifacts in the EEG signals and mitigate the vanishing gradient problem. The ReLU activation function is defined as $f(x) = \max(0, x)$, which means it sets any negative value of the feature to zero. Moreover, to handle the exploding gradients problem Batch normalization is utilized after each convolution layer to normalize the inputs to each layer within certain ranges (0,1). Batch normalization helps in maintaining stable gradients throughout the network. This leads to faster learning and better performance than other activation functions, such as Tanh [26]. After the final CL, the extracted features are flattened into a one-dimensional array (feature vector) by the flattening layer. The feature vector includes the most important EEG features extracted by the base model and serves as the feature extraction point in the base model. The training process, which incorporates the use of a 1D CNN, is trained for n = 100 epochs (iterations) with randomly initialized parameters and consists of the following steps:

- *Extraction:* The trained 1D CNN extracts features from the time and frequency domain signals automatically.
- *Classification:* The 1D CNN predicts the class for each input record.
- *Evaluation:* The performance of the trained model is evaluated using accuracy, specificity, and sensitivity metrics by comparison.

After training the proposed base model, the truncated model is used (having removed the last layer of the 1DCNN model) to extract time and frequency domain vector data from the flattened layer while retaining the same training weights generated by the base classifier. The proposed deep features are then saved into a related file for further analysis.

2) Time Domain Deep Features (TD-DF): After the truncated model is computed, the proposed framework calculates the statistical properties such as standard deviation, root mean square, mean, mean absolute deviation, and peak estimation of the deep feature, as shown in Table II for further analysis and interpretation of the rule generation. The peak estimation, excluding features containing zeros, is estimated to capture subtle feature variations. To facilitate feature preservation, the data is smoothed and filtered using a Savitzky-Golay filter having $window\_length = 5, polynomial\_order = 2$, and a Butterworth low-pass filter has $cutoff\_freq$= 0.1, $filter\_order$ = 3, and $Nyquist\_freq$= 0.5. Equations (2) and (3) present the Savitzky-Golay and Butterworth low-pass filters, respectively. Fig. 3 illustrates the filter process of the deep feature.

TABLE II
EQUATIONS OF THE STATISTICAL PROPERTIES OF THE EEG SIGNALS

Time domain	Frequency domain
$\text{time}_{\text{mean}} = \frac{1}{N}\sum_{i=1}^{N} x_i \quad (4)$	$\text{freq}_{\text{dev}} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i-\mu)^2} \quad (5)$
$\text{time}_{\text{dev}} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i-\mu)^2} \quad (6)$	$\text{freq}_{\text{skew}} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{F_i-\mu_f}{\sigma_f}\right)^3 \quad (7)$
$\text{time}_{\text{rms}} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} x[n]^2} \quad (8)$	$\text{freq}_{\text{mean}} = \frac{1}{N}\sum_{i=1}^{N} x_i \quad (9)$
$\text{time}_{\text{med}} = \frac{1}{N}\sum_{i=1}^{N} \lvert x_i-\mu\rvert \quad (10)$	$\text{freq}_{\text{var}} = \frac{1}{N-1}\sum_{i=1}^{N}(F_i-\mu_f)^2 \quad (11)$
$\text{time}_{\text{peak}} = \max_{i=1}^{N}\lvert x_i\rvert \quad (12)$	$\text{freq}_{\text{med}} = \begin{cases} F_{\frac{N+1}{2}}, & \text{if } N \text{ is odd} \\ \frac{F_{\frac{N}{2}}+F_{\frac{N}{2}+1}}{2}, & \text{if } N \text{ is even} \end{cases} \quad (13)$

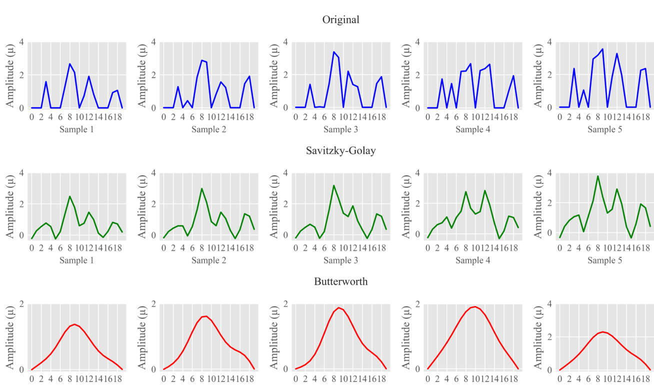

Fig. 3. The original signals, Savitzky-Golay and Butterworth low-pass filters for peak estimating from extracted features. The top row (blue color) shows the original unfiltered data across five different samples. The middle row (green color) demonstrates the Savitzky-Golay filter, which smooths the signal while preserving features like the height and width of peaks and removing any ripples. The bottom row (red color) illustrates the result of applying a Butterworth filter and shows its impact on the signal with a different smoothing effect that attenuates higher frequencies. Each column corresponds to a unique sample, illustrating the variance of the feature data and the effects of each filtering method.

$$y_i = \frac{1}{M}\sum_{j=-k}^{k} c_j x_{i+j} \tag{2}$$

$$H(s) = \frac{1}{\sqrt{1+\left(\frac{s}{\omega_c}\right)^{2n}}} \tag{3}$$

Let y_i be the smoothed signal at point, x_{i+j} the original input data at end $(i+j)$, i ranges from k to $N-k-1$ (N: total number of data points), j ranges from $-k$ to k (this creates a window of size $2k+1$), c_j are the filter coefficients and M is a normalization factor, y is the original input data, i ranges from m to $N-m-1$, while $H(s)$ represents the frequency response, ω_c is the filter's cutoff frequency and the stop band (the range of frequencies attenuated), n is the filter order, and s is the complex frequency variable, where σ is the fundamental part (attenuation) and $j\omega$ is the imaginary part (angular frequency). The Savitzky-Golay filter is utilized primarily for its ability to preserve important signal characteristics, such as height and width, and signal trends. On the other hand, the Butterworth low-pass filter is known for its frequency response, which is primarily used to allow lower-frequency signals to pass while attenuating higher-frequency noise.

3) Frequency Domain Deep Features (FD-DF): To generate comprehensive rules for each class (non-seizure (NZ), seizure free or transition state (TS), and seizure (SZ)), relying solely on TD-DF, which may not provide sufficient information. Therefore, for accurate ES detection, it's important to incorporate FD-DF. We compute the statistical properties of the vectors data of the time and frequency domain, including standard deviation, variance, mean, skewness, and median, as shown in Table II. These statistical computations aid in creating a distinctive set of attributes that encapsulate the primary characteristics of the signals' frequency. These features are subsequently used for further analysis and interpretation of the rules for each EEG class.

4) Multi-Domain Deep Features (MD-DF): We combine the statistical properties of both TD-DF and FD-DF to establish an MD-DF analysis. This consolidated information is stored in a related file directory, further use rule generation. Fig. 4 depicts histograms of the TD-DF and FD-DF. The first row demonstrates the distribution of various TD-DF, including the mean, absolute deviation, mean, peak estimation, root mean square, and standard deviation. All these statistical time-deep features exhibit symmetrical and normal distributions. The second row displays the distribution of FD-DF, including mean, median, skewness, standard deviation, and variance. The plot reveals that the median, standard deviation, and variance display a right-skewed distribution, whereas skewness presents a left-skewed distribution. After visualization and processing, we combine MD-DF to have both views form a multi-view deep feature, all the processes from DF extraction to integration as shown in Fig. 5. After the integration of multi-view deep features (MV-DF), we employ principal component analysis (PCA) for feature selection and dimensional reduction. PCA identifies the directions, known as principal components, in the feature space where the data exhibits the most variance. Each of these principal components represents a linear combination of the original MV-DF. Our approach entails focusing on the principal components that account for the majority of the variance. The corresponding weights of the MV-DF in these top-ranking components serve as indicators of their relative importance.

In the previous sections, we discussed the construction of MV-DF from 1D CNN and the statistical properties of the time and frequency domain deep features. The integration of diverse deep features (DF) with multi-view learning (MV-L) necessitates the formulation of a multi-view framework, hence The proposed study implement MV-F classifier, which includes a random forest classifier for each view. This classifier uses algorithms that can discern individual views within the data. In this study, the MV-F classifier generates multiple bootstrap

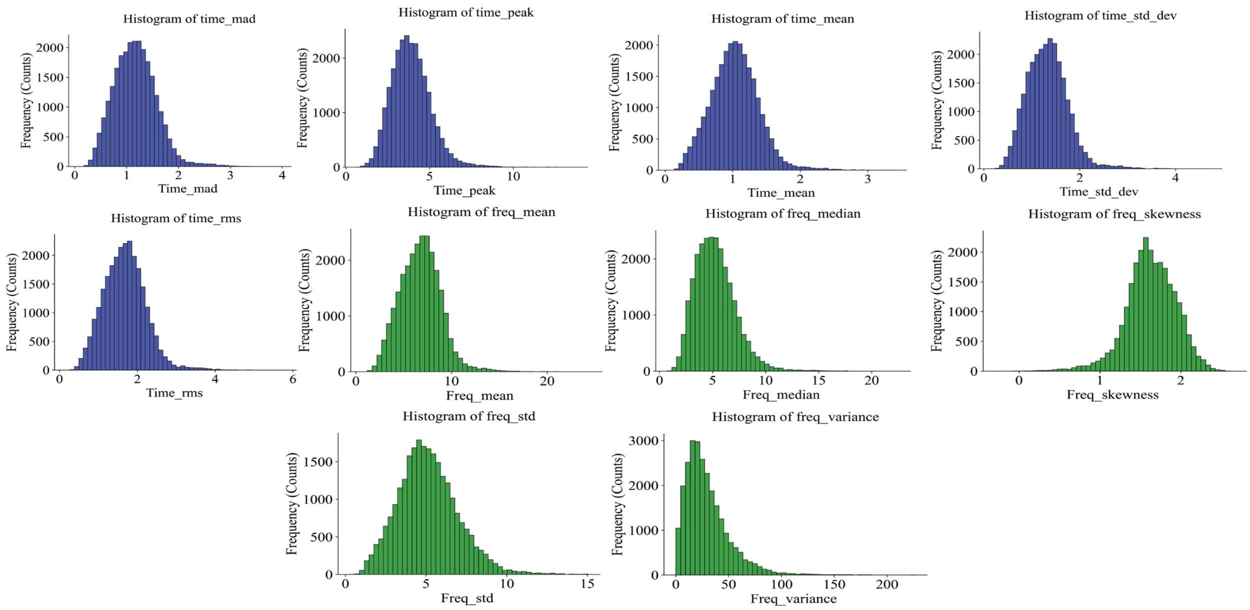

Fig. 4. The distribution of multi-domain deep feature properties. Each histogram corresponds to a different feature extracted, with time-domain features shown in blue and frequency-domain features in green.

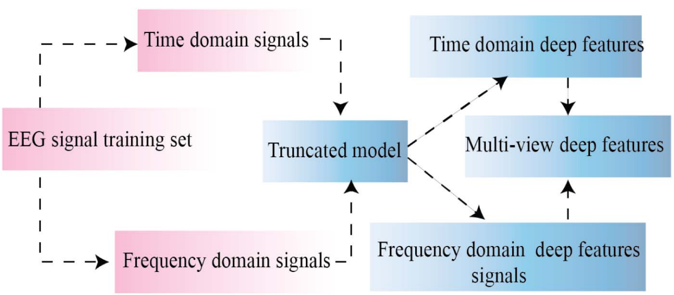

Fig. 5. The framework for multi-view deep feature extraction from the base model. The integrated approach leverages the complementary characteristics of each domain to enhance the feature representation.

samples from the original MV-DF training data, where instances are randomly selected with replacement. Decision trees are built from each bootstrap sample, selecting the best feature for splitting the data at each node from a random subset of features. This construction proceeds until certain criteria are met, such as a maximum $tree\_depth = 10$ or a minimum number of instances in a $leaf\_node = 10$.

The MV-F algorithm compiles rules by combining the output of all decision trees within the ensemble. For classification, a majority vote approach is typically adopted, where each tree's prediction counts as one vote, determining the final prediction by the highest vote count. Individual views are processed with separate forest models to derive classification rules for the SZ, NS, and TS classes. The integrated results from each view's forest model form a comprehensive ensemble of forest classifiers. The efficacy of the classification at this stage is measured by the rules generated by each model in the ensemble, as depicted in Fig. 6. Practically, any MV-F model can be utilized for EEG-based epilepsy detection, applying the MV-DF developed through this described method. It's critical

Algorithm 1: Multi-View Forest Classifier (MV-F).

Require: Multi-view deep features (MV-DF), $X = \{X_1, X_2, \ldots, X_n\}$
Require: Training labels $Y = \{y_0, y_1, y_2\}$, show Non-Seizure, Transition, and Seizure.
Require: Number of layers L

Initialize MV-F with L layers
for $I = 1$ to n **do**
  for $l = 1$ to L **do**
    if $l = 1$ **then**
      Train cascade forest f_l^i using view i and layer l
      Obtain class vector R_l^i for view i and layer l
    else
      Concatenate $R_1(l-1), R_2(l-1), \ldots, R_n(l-1)$
      Train cascade forest f_l^i using view i and layer l
      Obtain class vector R_l^i for view i and layer l
    end if
  end for
end for
for $l = 1$ to L **do**
  Concatenate $R_l^1, R_l^2, \ldots, R_l^n$, as input for layer l
  $R_l^i = R_{y_0}^{i,l}(X_i) + R_{y_1}^{i,l}(X_i) + R_{y_2}^{i,l}(X_i)$
end for
Train the final classifier using concatenated class vectors
$Rules_i \leftarrow$ Generate Rules (MV-DF Classifier)
Ensure: Multi-view forest classifier

to recognize that MV-F is optimally utilized as part of the proposed MV-DF method for epilepsy detection. Additionally, algorithm 1 explains the main process of MV-F for rule generation and classification of non-seizure, transition, and seizure. Where $X = \{X_1, X_2, \ldots, X_n\}$ represents the

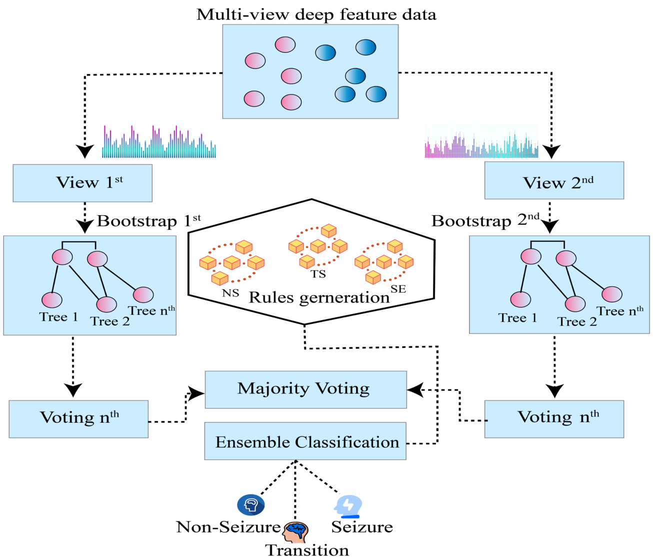

Fig. 6. The architecture of the proposed model indicates a multi-view approach to feature analysis. Initially, multi-view deep features are segmented into two separate views. View 1 and View 2 correspond to different perspectives or feature sets received from selected features.

multi-view deep feature training data and $Y = \{y_1, y_2, y_3\}$ shows the labels. The MV-F classifier consists of L layers, and each layer consists of a cascade random forest for each view, denoted as $F_l^i(X_i)$, where l is the layer index and i is the view index. Moreover, Algorithm 2 presents the main process of the framework, from deep feature extraction to classification, including the three main components from B to C of the proposed AMV-DFL framework explained in each section.

C. Construction of Interpretability and Explainability of the Proposed Model

After generating the rules using the MV-F model as discussed in the previous section, understanding the significance of deep feature analysis in rule generation for different classes is crucial for understanding the model's predictions. SHAP (Shapley Additive Explanations) enhances the interpretability of machine learning outcomes by attributing importance values to individual features. These values offer both local and global explanations, shedding light on the influence of each feature on the model's predictions. By quantifying the contributions of deep features to the rules generated by the MV-F model, SHAP facilitates a more in-depth comprehension of the decision-making process. The integration of SHAP into the MV-F model not only improves interpretability but also aids in the development of rules for epileptic seizure (ES) detection, thereby supporting both validation and potential enhancements of the model. This improved clarity is instrumental for advancing epilepsy detection systems, providing benefits for medical practitioners and patients. TreeSHAP, part of the SHAP framework, is specifically designed for tree-based machine learning models. It employs Shapley values to determine the contribution of each feature to individual predictions, offering a consistent and locally accurate measure

Algorithm 2: Multi-View Deep Feature Learning (MV-DFL).

```
procedure multi-view deep feature, TMV − F, deep_feature
    Train truncated model ← Train 1D CNN (TMV − F)
    Initialize com_DFi ← Empty list
    for i ∈ {1, ..., N deep feature} do
        deepTF_i ← deep_feature(Train truncated model)
        deepFF_i ← deep_feature(Train truncated model)
        com_DF_i ← fuse features(deepTF_i, deepFF_i)
    end for
    MV-F classifier ← forest classifier(com_DF_i)
    Rules_i ← generate rules (MV-F classifier)
    F_class ← Integ_rule({Rules_i}deep_feature)
    Initialize y_0 ← 0, y_1 ← 1, y_2 ← 2,
    for each test data in test dataset do
        prediction ← predict y_2 (F_class, test data)
        if prediction == TRUE then
            Calculate the y_2
        else
            Calculate y_0, y_1
        end if
    end for
end procedure
```

of feature importance. The TreeSHAP model is described mathematically by (14):

$$\Phi_i = \sum_{S \subseteq \{x\} \setminus \{i\}} \frac{|S|!(|\{x\}| - |S| - 1)!}{|\{x\}|!} [f_{S \cup \{i\}}(x) - f_S(x)] \tag{14}$$

where the Φ_i denotes the Shapley value for feature i, S represents all subsets of features excluding feature i, $f_{S \cup \{i\}}(x)$ is the model's prediction when trained on subset S plus feature i, and $f_S(x)$ is the prediction when the model is trained only on subset S of features.

D. Experimental Setup and Performance Indices

The study employs the Bonn EEG dataset experimental type (AB, CD, E). The set (AB) is labeled as 0 and shows non-seizure (NS), while the set (CD) presents the seizure-free or transition state (TS) as labeled 1 and the set E indicates the seizure (SZ) as labeled 2. Moreover, to balance the seizure file (set E), we apply the Synthetic Minority Oversampling Technique (SMOTE) technique. To evaluate the performance of the framework, a 10-fold cross-validation is performed. Moreover, we utilize grid search to identify the optimal hyperparameters for the base model and the proposed MV-Forest classifier by systematically evaluating model performance across various hyperparameter combinations. The hyperparameters include the number of filters, batch size, optimizer, kernel size, stride, and learning rates for the base classifiers. For the proposed classifiers, we employ hyperparameters such as the number of estimators (n_estimators), maximum depth (max_depth), minimum sample split (min_samples_split), minimum sample leaf

TABLE III
THE BONN EEG DATASET INCLUDED DIFFERENT EEG STATES OF VARIOUS PATIENTS

Case	Patient State	Class	Electrode Type	Number of files	Sample duration (sec)
Set A	Eyes opened	Healthy (non-seizure)	Surface	100	23.6
Set B	Eyes closed	Healthy (non-seizure)	Surface	100	23.6
Set C	Seizure free or Transition	Transition	Intracranial	100	23.6
Set D	Seizure free or Transition	Transition	Intracranial	100	23.6
Set E	Seizure (ictal-period)	Seizure	Intracranial	100	23.6

TABLE IV
PERFORMANCE OF THE TRADITIONAL MULTI-VIEW FEATURES USING THE PROPOSED APPROACH WITH BASELINE METHODS

Features	Methods	Acc (%)	Prec (%)	Sens (%)	Spec (%)	F1-score (%)	Time complexity (sec)
Time domain feature	KNN	80.30	79.40	79.25	79.10	79.15	5.09
	NB	82.32	80.32	80.25	78.39	79.40	4.09
	DT	83.31	83.45	83.41	82.04	82.50	8.56
	SVM	82.14	81.20	81.10	81.30	81.20	9.30
	Proposed method	92.98	92.90	92.80	92.30	92.40	8.45
Frequency domain features	KNN	80.10	79.29	79.20	78.50	78.90	8.10
	NB	84.19	83.68	83.60	83.10	83.40	6.36
	DT	86.09	84.35	84.30	84.20	84.15	11.81
	SVM	83.70	82.10	82.20	82.10	82.10	10.10
	Proposed method	93.10	93.04	93.05	92.90	93.05	10.12
Time-frequency domain features	KNN	83.15	83.20	83.13	83.10	83.11	11.34
	NB	88.90	88.18	88.10	86.20	87.30	10.45
	DT	92.19	91.75	91.70	90.05	91.35	10.10
	SVM	89.89	88.80	88.90	88.50	88.50	11.30
	Proposed method	97.40	97.36	97.30	97.04	97.15	10.12

TABLE V
THE PERFORMANCE OF THE MULTI-VIEW DEEP FEATURE USING THE PROPOSED MODEL WITH BASELINE METHOD

Features	Methods	Acc(%)	Pres (%)	Sens (%)	Spec (%)	F1-score (%)	Time complexity (sec)
Time domain deep feature	KNN	85.70	85.60	84.50	84.90	84.70	7.09
	NB	88.80	87.90	87.80	87.30	87.50	6.89
	DT	90.98	89.50	89.45	88.10	88.30	10.31
	SVM	89.96	88.30	88.23	87.67	87.90	10.83
	Proposed method	95.98	95.85	95.80	95.30	95.50	10.09
Frequency domain deep features	KNN	86.20	86.18	86.10	85.50	85.80	10.23
	NB	90.89	90.66	90.60	90.01	90.30	8.90
	DT	93.40	93.18	93.10	93.02	93.10	10.50
	SVM	91.10	91.05	90.12	90.09	89.12	10.83
	Proposed method	96.90	96.21	96.10	96.30	96.20	10.29
Time-frequency domain deep features	KNN	90.90	89.19	89.10	89.05	89.05	11.06
	NB	93.79	93.25	93.20	93.40	93.30	11.45
	DT	96.10	93.88	95.80	95.10	95.40	12.34
	SVM	93.13	93.10	92.14	92.19	91.13	11.83
	Proposed method	99.40	99.20	99.10	99.10	99.15	11.15

(min_samples_leaf), and bootstrap, all of which are determined through grid search. Therefore the best parameters for the base model (1D CNN) are the number of filters (24, 16, 8), kernel size (5, 2, 3, 3), and stride (2,2,2) for different layers of conventional while 100 epochs, batch size=32,optimizer='adam' and learning rate is 0.001 with categorical cross-entropy loss function are used in the proposed study. Moreover, the best hyper-parameters for the multi-view forest (MV-F) classifier are n_estimator = 100, max depth = 10, min_samples_split is 5, min samples leaf is set to 10, bootstrap is set to 'True'. On the other hand, the proposed study evaluates different models, such as support vector machines (SVM), K-Nearest Neighbors Algorithms (KNN), Naive Bayes classifier (NB), and decision trees (DT), to evaluate the performance of the rules-based classifier. The SVM kernel is selected as linear. We determine peak information from the selected features; any feature with a value of 0 is initially excluded. For peak estimation within the refined feature vectors, we utilize the find_peaks function from the SciPy'signal processing library. The function counts the number of peaks by examining adjacent features to locate all the local maxima in the feature vector. The maximum amplitude contains a peak if it has a high value and is significantly higher than its neighboring maximum amplitude value. In the initial experimental study, it was observed that the raw EEG data and the extracted features have distinguishable peak patterns. Hence, the process of peak estimation is employed to encapsulate the characteristics of the features, aiming to the rules and explanations. The experiments were carried out using an Intel Core i7 CPU and a GPU, ensuring efficient processing and analysis of the data. For model development and training, libraries such as TensorFlow, Pandas, Keras, NumPy, and Scikit-Learn were utilized within the Python 3.8 environment. The language used to complete the projects is Python 3.7, utilizing various libraries for implementation. The performance of the proposed classifier is evaluated using five common performance indices: accuracy (Acc), precision (Prec), sensitivity (Sens), specificity (Spec), and F1-score (F1). These

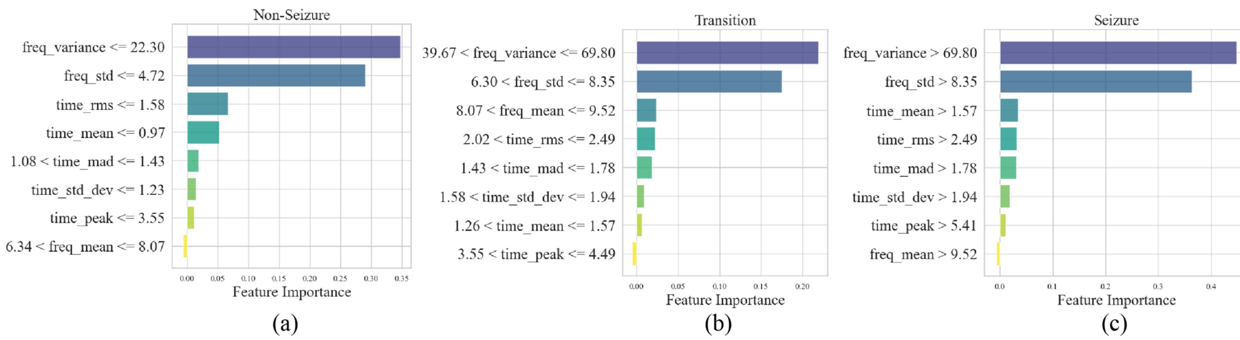

Fig. 7. Represents the importance of the deep feature and its effect on the outcomes of the proposed model. The y-axis shows that the deep features have specific values for the rules, while the x-axis shows the feature importance value. (a) The generated rules for the non-seizure, (b) transition, and (c) for the seizure.

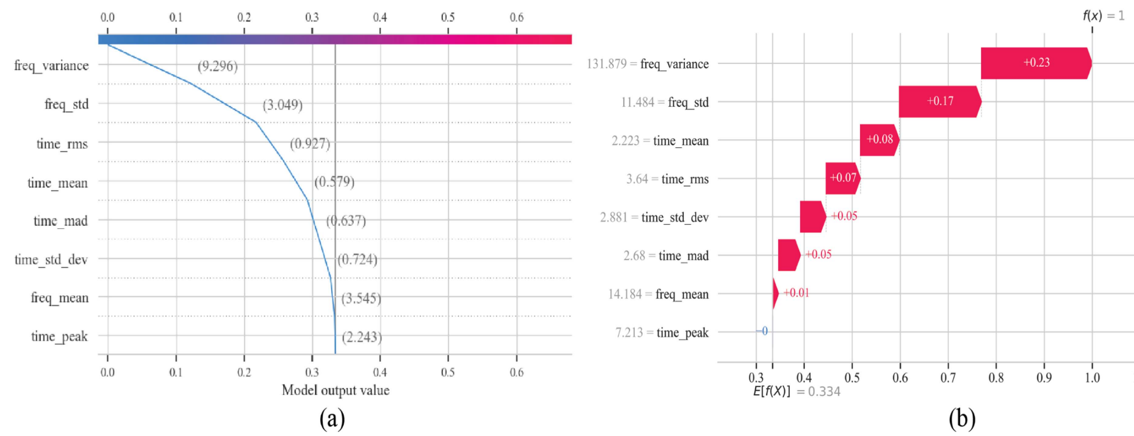

Fig. 8. Visualization of the waterfall plot of the proposed model output. The subfigure (a) shows a step-by-step decrease in model output value, indicating the contribution of each feature to the reduction in the final detection score. The subfigure (b), the x-axis represents the expected value (E[f(X)]) of the model's output before considering the impact of each feature, while the y-axis lists the features in order of their absolute impact on the model's output.

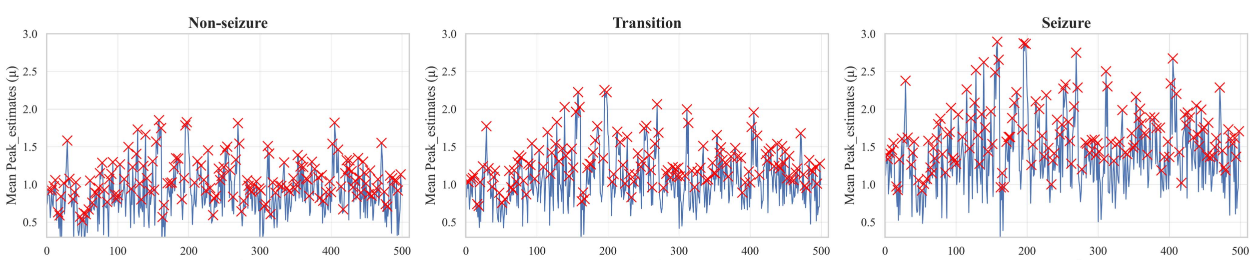

Fig. 9. Visualization of the mean peaks_estimation of different classes after detection. Each panel represents one of the classes, and visualization of the variation of the $mean\_peak$ using 500 samples. The red crosses indicate the estimated peak values for each sample, and the blue line represents the mean of these estimates, illustrating the fluctuation within each class.

metrics provide a comprehensive evaluation of the system's performance. The equations and explanation of the above are in the given literature [5], [6], [7].

IV. Experimental Results and Discussion

A. Data Acquisition

The proposed study utilizes a public EEG signal database from Bonn University, Germany, consisting of five sets, A to E, each with 100 single-channel EEG signal recordings [24]. The sampling frequency is 173.61 Hz with a 12-bit A/D resolution, each lasting 23.6 seconds. Sets A and B contain EEG signals from five healthy normal, awake, and relaxed subjects, representing non-seizure activity. Sets C, D, and E consist of EEG data from five epileptic patients. Set C and Set D are from the Seizure free or transition state of the seizure patients, while Set E presents the seizure activity of the patients. Table III shows the description of the Bonn EEG datasets,while in the preprocessing of the EEG data, a Butterworth filter is used, which includes a low cutoff frequency of 0.1 Hz, a high cutoff frequency of 40 Hz,

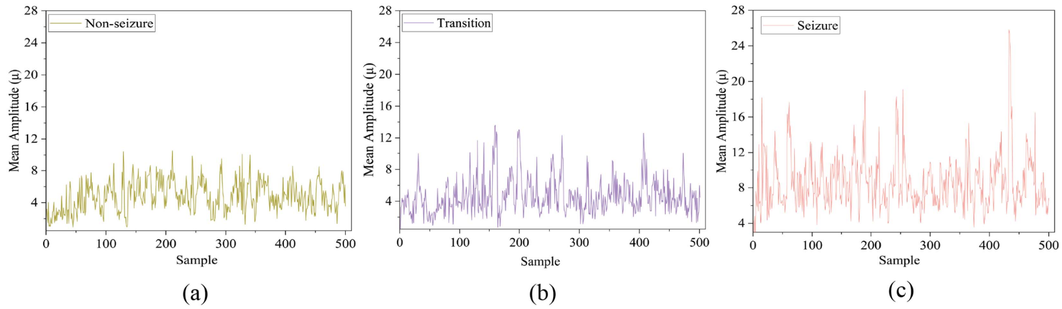

Fig. 10. Visualization of the different classes after seizure detection. Subfigure (a) shows the non-seizure state, characterized by relatively stable and low-amplitude fluctuations indicative of normal brain activity. Subfigure (b) represents the transition state, where there is a noticeable increase in amplitude variability, suggesting the onset of abnormal neural activity. Subfigure (c) illustrates the seizure state, which is markedly more volatile, with significant spikes in amplitude, reflecting the intense and abnormal activity of the brain.

a filter order of 3. Moreover, for normalization, we applied a 'standard scaler' to normalize the EEG data to improve the performance of the model.

B. Analysis and Effectiveness of Multi-View Deep Feature With Explainable Artificial Intelligence (XAI)

The evaluation presented in this study focuses on analyzing the effectiveness of multi-view deep features (MV-DF) for EEG ES detection. Specifically, we compare all the sibling methods' performance with the proposed model mentioned above using the traditional multi-view feature (TMV-F) as summarized in Tables IV and V. Upon analysis, all classifiers demonstrate favorable performance when utilizing deep features (DF). Using DF in different sibling models outperforms traditional features, yielding higher accuracy, precision, sensitivity, and specificity in ES detection. Moreover, the proposed model has the same time complexity (sec) as other sibling ML models, while among different deep feature domains, TD-DF shows the poorest performance and FD-DF demonstrates moderate results. Notably, MV-DF exhibits excellent performance because it provides a more comprehensive view of the signal offering a rich representation of the EEG data. The findings of the proposed study have significant implications in the field of EEG ES detection. The proposed MV-F approach enhances the performance of classifiers for ES detection by 5% accuracy using TMV-F and 4% MV-DF as compared to other sibling models. The improved accuracy, sensitivity, and specificity of the classifiers can contribute to more accurate and reliable ES detection, aiding in timely diagnosis and treatment decision-making. Moreover, to check the proposed model outcomes based on rule generation utilizing MV-DF, and explainability is conducted by XAI. Fig. 7 presents the importance and effectiveness of the deep features for the proposed model. They assist in determining a feature's significance within the model. Larger SHAP values for features have a more significant influence on the model's detection. For the NS class, the most important features are variance, standard deviation, and mean while for the TS and SZ classes, the best features are variance, standard deviation, and root mean square. These deep features play a crucial role in effectively distinguishing between different EEG classes. Fig. 10 shows the

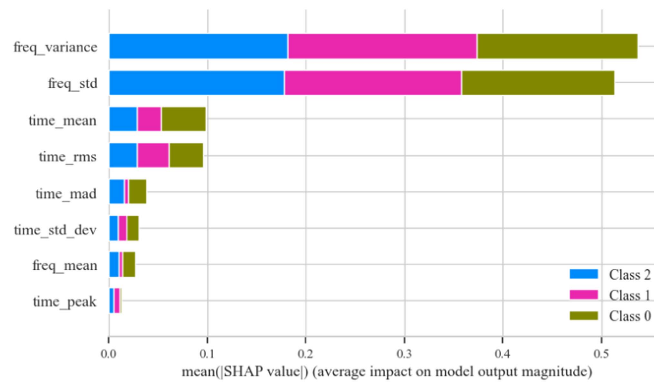

Fig. 11. The summary plot of the proposed model output. The different colors present the contribution of the feature in the decision-making process. The x-axis shows the importance of the feature and the y-axis depicts the mean SHAP values after seizure detection.

predicted samples for the NS, TS, and SZ classes. The analysis reveals that SZ samples exhibit higher amplitude compared to transition and non-seizure samples, while TS samples show a normal amplitude and NS show a low amplitude compared to TS samples. The analysis also shows that SZ classes have high peak estimation values, the TS classes have moderated, and the NS classes have a low peak estimation value, as shown in Fig. 9.

Furthermore, Fig. 8(a) presents the decision plot (DP) of individual features that contribute to a model's detection, for instance, 0. The x-axis is centered around the explainer's expected value (EV), similar to how a linear model's effects are referenced to its intercept. Each SHAP value is thereby associated with the model's predicted value. The y-axis of the plot corresponds to the model's features. In addition, Fig. 8(b) shows the waterfall plot (WP) for the proposed model and shows the response patterns for TP, FP, FN, and TN. The blue and red bars in the WP display individual features and their respective contributions to the cumulative classification score. These contributions can either augment or reduce the final score, demonstrating the impact of each feature on the model's decision-making process. Subsequently, Fig. 11(b) shows the Summary Plot (SP) of the MV-F model, which serves as a Visualization Plot (VP) for a global overview. The plot arranges all model features on

TABLE VI
RULES GENERATED BY THE PROPOSED MODEL FOR THE NON-SEIZURE CLASS

No. of Rules	Rules generation	Classification
1	If the $freq_{var} \leq 22$, the $time_{rms} \leq 1.58$, and the $freq_{mean} \leq 6.35$	Non-seizure
2	If the $freq_{var} \leq 20$, the $time_{rms} \leq 1.50$, and the $time_{mean} \leq 1.26$	Non-seizure
3	If the $freq_{var} \leq 21$, the $time_{rms} \leq 1.58$,the $time_{dev} \leq 3.55$, and the $freq_{dev} \leq 4.73$	Non-seizure
4	If the $freq_{var} \leq 19.24$, the $time_{rms} \leq 1.56$, the $time_{peak} \leq 3$, and the $time_{mean} \leq 1.10$	Non-seizure
5	If the $freq_{var} \leq 20.20$, the $time_{rms} \leq 1.58$, the $time_{rms} \leq 1.50$, and the $freq_{dev} \leq 6.30$, and the $time_{peak} \leq 3$	Non-seizure
6	If the $time_{rms} \leq 1.65$, the $freq_{mean} \leq 5.35$, $time_{mad} \leq 1.02$, the $time_{peak} \leq 3$, and the $time_{dev} \leq$ 3.43	Non-seizure
7	If the $freq_{var} \leq 22$, the $time_{rms} \leq 1.65$, the $time_{mean} \leq 1.05$, $time_{mad} \leq 1.24$, the $freq_{mean} \leq$ 6.24, $time_{dev} \leq 3$, the $time_{peak} \leq 4$, and $freq_{dev} \leq 6.30$	Non-seizure

TABLE VII
RULES GENERATED BY THE PROPOSED MODEL FOR THE SEIZURE -FREE OR TRANSITION CLASS

No. of Rules	Rules generation	Classification
1	If the $freq_{var}$∈[22.37, 39.66], thefreq˙dev∈ $[4.73, 6.30]$, and the $time_{mad} \in [1.08, 1.43]$	Transition
2	If the $freq_{var} \in [25.24, 39.10]$, the $time_{mad} \in [1.10, 1.45]$, and the $time_{peak} \in [4, 6]$	Transition
3	If the $freq_{var} \in [23.20, 38.18]$, the $time_{mad} \in [1.20, 1.43]$, the $time_{peak} \in [5, 6]$, and the $time_{dev}$ $\in [1.27, 1.58]$	Transition
4	If the $freq_{var} \in [23.40, 39.10]$, the $freq_{dev} \in [6.30, 8.38]$, and the $time_{mean} \in [1.27, 1.56]$	Transition
5	If the $freq_{var} \in [25, 39]$, $time_{mad} \in [1.20, 1.45]$, and the $time_{rms} \in [1.59, 2.34]$	Transition
6	If the $freq_{var} \in [25, 39]$, $time_{mad} \in [1.20, 1.45]$, $time_{peak} \in [4, 5]$, and $freq_{dev} \in [5.10, 6.30]$	Transition
7	If the $freq_{var} \in [22.37, 39]$, the $freq_{dev} \in [4.73, 6.30]$, the $time_{mad} \in [1.08, 1.43]$, the $time_{peak}$ $\in [4, 6]$, the $time_{mean}\in [1.26, 1.57]$, the $time_{dev} \in [5.10, 6.30]$, and the $freq_{mean} \in [6.35, 7.50]$	Transition

TABLE VIII
RULES GENERATED BY THE PROPOSED MODEL FOR THE SEIZURE CLASS

No. of Rules	Rules generation	Classification
1	If the $freq_{var}$ >39.50, the $freq_{dev} > 8.38$, and the $time_{mean} > 1.57$	Seizure
2	If the $freq_{var} > 40$, the $freq_{dev} > 9$, the $time_{mean} > 1.80$, and the $time_{rms} > 2.49$	Seizure
3	If the $freq_{var} > 39.90$, the $freq_{dev} > 9.70$, the $time_{mean} > 1.60$, and $time_{mad} > 1.60$	Seizure
4	If the $freq_{var}> 41.7$, the $time_{dev} > 1.94$, and the $time_{peak} > 6$	Seizure
5	If the $freq_{var}$ >42.50, the $freq_{dev} > 8.50$, the $time_{peak} > 7$, and the $freq_{mean} > 9.52$	Seizure
6	If the $freq_{var} > 41$, the $freq_{dev} > 10$, the $time_{rms} > 2.49$, and the $time_{peak} > 7$	Seizure
7	If the $freq_{var} > 40$, the $freq_{dev} > 8.60$, the $time_{mean} > 1.60$, the $time_{rms} > 2.49$, the $time_{mad} >$ 1.79, the $freq_{mean} > 9.52$, the $time_{dev} > 1.94$, and the $time_{peak} > 6$	Seizure

an x-axis, with the most influential ones occupying the upper sections and the least impact ones positioned at the bottom. In essence, the SP provides an overall perspective of feature significance throughout the entire model. Features with larger absolute SHAP values, indicating a stronger average effect on the model's predictions, are very important.

C. Automatic Rules Generation and Explanation

The study demonstrates the successful development of an approach for autonomously extracting high-quality deep features from flatten layer of the base model that have to include non-seizure, transition, and seizure records. These multi-view deep features were then employed to create a set of explanation rules that can effectively classify and explain the extracted NS, TS, and SZ feature records, as shown in Tables VI, VII,VIII. The generated rule by the proposed model represents a specific combination of feature values that contribute to the accurate classification and identification of each class pattern. These rules offer interpretability and explainability to the proposed model by revealing the key factors that distinguish SZ samples from other classes. By understanding the relationship between certain feature values and SZ classification, clinicians and researchers can gain insights into the underlying characteristics and markers of SZ. The presence of these rules enhances the transparency of the decision-making process and facilitates the identification of SZ patterns based on distinct amplitude characteristics. The proposed MV-F is bench-marked against recent models in the literature that utilize traditional features and multi-domain features with a single view. The proposed framework employs multi-view learning and often outperforms single-view learning in terms of prediction accuracy and other performance metrics because it has access to more diverse and complementary information. Moreover, the performance of the proposed method is also compared with the base classifier, 1D CNN. The proposed method performs better than the state-of-the-art methods that employ multi-domain and outperforms most alternative approaches and the base model, as shown in Table IX. A key advantage of the proposed framework is that it eliminates the need for domain experts to extract features from raw EEG data and generate autonomous rules. The generated rules demonstrate the effectiveness of the proposed model in accurately classifying each class sample. The interpretability provided by these rules enhances the transparency of the model's decisions, enabling

TABLE IX
COMPARATIVE RESULTS OF THE PROPOSED FRAMEWORK WITH RECENT STATE-OF-THE-ART APPROACHES

Publication	Approach	Features	View learning	Acc(%)	Spec(%)	Sens(%)	F1-score (%)	Time complexity (sec)
[22]	Multi-view deep learning	DF	Multi-view	88.90	94.10	-	-	-
[28]	PCA-RBFNN	TMDF	Single view	97.10	-	-	-	-
[29]	Filtering + FC-LSTM	TF	Single view	96.00	95.10	95.10	95.10	-
[30]	DWT + RF	TMDF	Single view	97.10	97.04	97.03	97.03	-
[13]	Enco+Deco	UMDF	Multi-view	98.30	98.16	-	-	-
[18]	SVM-DT,SVM-RF	UMDF	Multi-view	98.15	98.50	-	-	-
This study	Base Classifiers	DF of Base Classifier	Single view	98.10	98.50	98.10	98.30	60.67
This study	AMV-DFL Approach	TMV-F, MV-DF	Multi-view	99.40	99.20	99.10	99.15	11.15

TMDF, traditional multi-domain features; PCA-RBFnn, Principal Component Analysis-Radial Basis Function Neural Network.

clinicians to gain valuable insights into the features contributing to SZ classification. The findings highlight the potential of the proposed model for improving diagnostic accuracy in SZ detection and advancing our understanding of SZ characteristics. Overall, the proposed model exhibits excellent potential as an efficient and accessible solution for detecting each state in EEG data. Moreover, all the rules generated by the proposed model for each class (non-seizure (NS), transition (TS), and seizure (SZ) using patients-specific EEG datasets. In future work, more data sets need to be involved to improve the generalization ability of the model in EEG epileptic seizure detection.

V. CONCLUSION

This study presents an advanced multi-view deep feature learning framework based on machine learning approaches for enhancing ES detection using EEG signals. Initially, the TMV-F is employed to train the base classifiers of the 1D CNN model and the MV-DF used by the proposed MV-F model. In contrast, T-XAI was used to visualize the rules of interpretation and describe the proposed model generated by non-seizure, transition, and seizure classifications. The results highlight the importance of deep feature and multi-view learning for efficient epilepsy detection through EEG signals. Comparative experiments show the advantages of the examined deep features over traditional features with different state-of-the-art methods, while the proposed framework yields promising results, discovering novel biomarkers.

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